

MODULE 1 ASSIGNMENT

HOW THE INTERNET DELIVERS A WEB PAGE

Course: PCCST501 – Computer Networks

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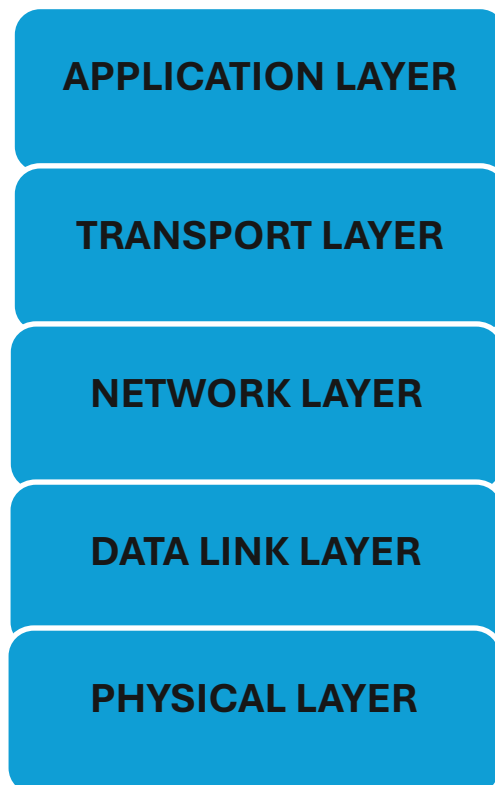
College: VJCET VAZHAKULAM

Date: 20/07/2026

Introduction

For this assignment, **Google** (www.google.com) has been selected as the website to demonstrate how the Internet delivers a webpage. Whenever a user enters www.google.com into a web browser and presses **Enter**, several networking protocols work together to retrieve and display the requested webpage.

LAYERS OF TCP/IP PROTOCOL



Application Layer: Provides network services to user applications using protocols such as HTTP, FTP, DNS, and SMTP.

Transport Layer: Ensures reliable end-to-end communication between devices using TCP or UDP.

Network Layer: Routes data packets across networks using IP addresses.

Data Link Layer: Transfers data between devices on the same local network using MAC addresses.

Physical Layer: Transmits raw bits over the physical communication medium such as cables or wireless signals.

Role of DNS in Converting the Domain Name into an IP Address

The **Domain Name System (DNS)** is often called the "**phonebook of the Internet.**" It converts human-readable domain names, such as www.google.com, into numerical **IP addresses** that computers use to identify and communicate with web servers. When a user enters a website address into a browser, the browser first sends a request to a DNS server. The DNS server searches its records and returns the corresponding IP address of the website. Using this IP address, the browser establishes a connection with the web server and requests the desired webpage. Without DNS, users would have to remember and enter numerical IP addresses instead of easy-to-remember domain names, making Internet browsing much more difficult.

HYPERTEXT TRANSFER PROTOCOL

The **Hypertext Transfer Protocol (HTTP)** is an application-layer protocol used for communication between a web browser and a web server. After the browser obtains the website's IP address through DNS, it sends an **HTTP request** to the web server asking for the required webpage. The server processes the request and returns an **HTTP response** containing the webpage's HTML, images, CSS, JavaScript, and other resources. The browser then interprets these files and displays the complete webpage to the user. Most modern websites use **HTTPS**, which encrypts the communication to ensure secure data transfer.

The **Transmission Control Protocol (TCP)** provides reliable communication by ensuring that data is delivered accurately and in the correct order between the web browser and the web server. Before sending HTTP data, TCP establishes a connection using the **three-way handshake**. It uses **sequence numbers** and **acknowledgements (ACKs)** to verify that all data packets are received correctly. If any packet is lost or corrupted during transmission, TCP automatically **retransmits** it. This reliability ensures that the complete webpage, including HTML, images, CSS, and JavaScript files, is delivered correctly to the user's browser.

Client-Server Architecture

Google follows the **Client-Server architecture**. In this model, the **web browser** acts as the **client**, and Google's computers act as the **server**. When a user enters www.google.com, the browser sends an HTTP request to Google's web server. The server processes the request, retrieves the required webpage, and sends an HTTP response back to the client. This architecture is suitable because the server stores and manages the website's data, while clients only request and display the information. Therefore, Google uses the **Client-Server architecture**, not the **Peer-to-Peer (P2P)** architecture.

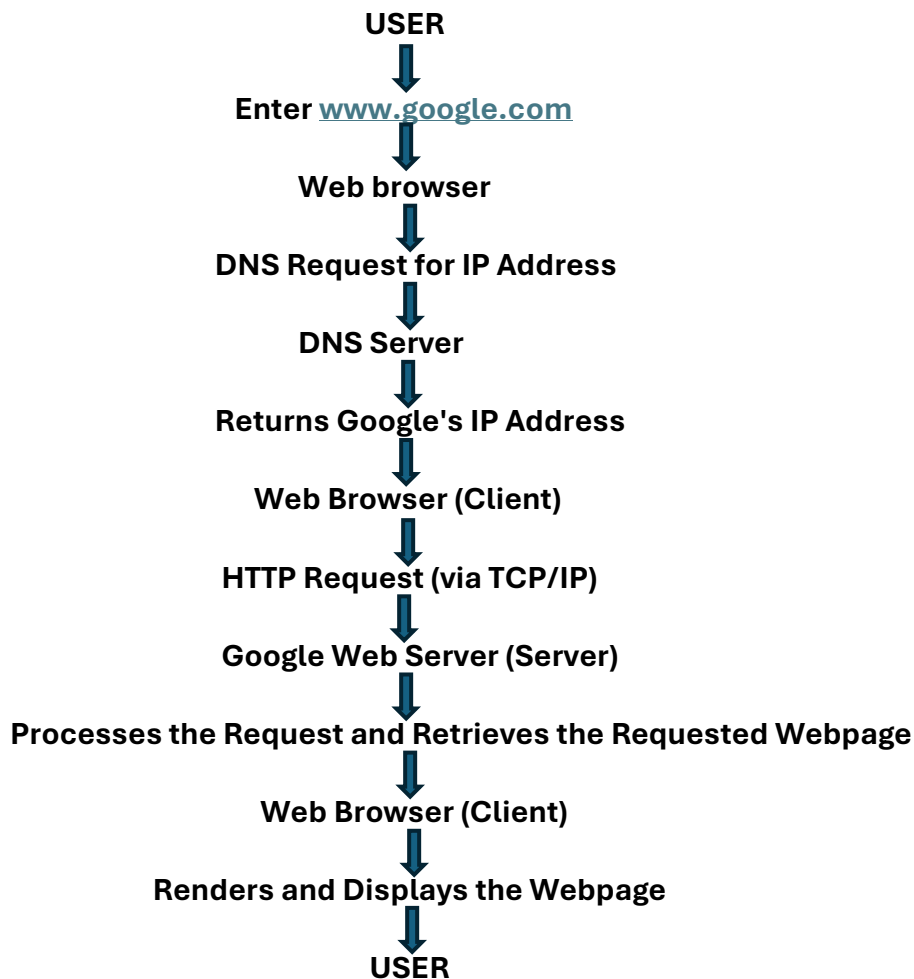
HTTP	FTP
• Hypertext Transfer Protocol	• FILE TRANSFER PROTOCOL
• Used to access and transfer web pages	• Used to upload and download files between computers
• Commonly uses Port 80 (HTTP) or Port 443 (HTTPS)	• Commonly uses Port 21 and Port 20
• Mainly used by web browsers to view websites.	• Requires an FTP client or FTP software for file transfer
• Transfers web content such as HTML, CSS, images, and JavaScript	• Transfers files such as documents, images, videos, and software

COOKIES

Cookies are small text files stored by a web browser on a user's device to remember information about their browsing session. They improve the user experience by remembering login details, language preferences, shopping cart items, and other personalized settings, so users do not have to enter the same information every time they visit a website.

Example: When a user logs in to Gmail and selects "Remember me", a cookie is stored in the browser. The next time the user visits Gmail, the website recognizes the cookie and keeps the user signed in, eliminating the need to log in again. This makes browsing faster, more convenient, and personalized.

FLOW DIAGRAM



CONCLUSION

Application-layer protocols play a vital role in enabling communication between users and Internet services. Protocols such as DNS, HTTP, and HTTPS work together to ensure that websites can be accessed quickly, efficiently, and securely. DNS converts domain names into IP addresses, HTTP and HTTPS enable the exchange of web pages between browsers and servers, and TCP ensures reliable data transmission. Cookies further improve the user experience by remembering user preferences and login information. Together, these technologies make everyday activities such as web browsing, online shopping, email, and video streaming simple and seamless. Understanding the role of application-layer protocols helps us appreciate how different networking components cooperate to deliver reliable, secure, and efficient Internet communication.